

10 MODBUS-RTU MODE

In MODBUS-RTU mode, the instrument sends the acquired measurements only if specifically requested by the PC, PLC or data logger.

The mode is available with RS232, RS485 and RS422 serial connections.

Communication parameters should be set in the PC or data logger as follows:

- Baud rate: same as the setting in the instrument (default = 19200)
- Data bits: 8
- Parity: same as the setting in the instrument (default = even)
- Stop bits: same as the setting in the instrument (default = 1)

MEASUREMENT READING (function **04h**)

The Modbus function code 04h allows reading the values measured by the instrument. The table below lists the available quantities with the corresponding register number:

TAB. 10.A - Input Registers

Register number	Quantity	Format	From FW version	Note
1	Wind speed (x100)	unsigned 16 bits	1.00	(1)
2	Wind direction in degrees (x10)	unsigned 16 bits	1.00	
3	Sonic temperature measured by the first transducers pair (x10)	16 bits	1.00	(1)
4	Sonic temperature measured by the second transducers pair (x10)	16 bits	1.00	(1)
5	Average of the two sonic temperatures measured by the two transducers pairs (x10)	16 bits	1.00	(1)
6	Temperature measured by Pt100 sensor (x10)	16 bits	1.00	(1)
7	Relative humidity in %RH (x10)	unsigned 16 bits	1.00	
8	Barometric pressure (x1000 if the unit of measurement is atm, x10 in the other cases)	unsigned 16 bits	1.00	(1)
9	Compass angle in degrees (x10)	unsigned 16 bits	1.00	
10	Solar radiation in W/m ²	unsigned 16 bits	1.00	
11	Mean wind speed (x100)	unsigned 16 bits	1.00	(1)
12	Mean wind direction in degrees (x10)	unsigned 16 bits	1.00	
13	Absolute humidity in g/m ³ (x100)	unsigned 16 bits	1.00	
14	Dew point temperature (x10)	16 bits	1.00	(1)
15	Wind direction in degrees (x10) with extended feature (see page 26)	unsigned 16 bits	1.00	
16	Wind speed (x100) along V-axis	unsigned 16 bits	2.00	
17	Wind speed (x100) along U-axis	unsigned 16 bits	2.00	

Register number	Quantity	Format	From FW version	Note
18	Status register bit0=1 ⇒ speed measurement error bit1=1 ⇒ compass measurement error bit2=1 ⇒ temperature meas. error bit3=1 ⇒ humidity measurement error bit4=1 ⇒ pressure measurement error bit5=1 ⇒ solar rad. measurement error	unsigned 16 bits	2.00	
19	Wind speed unit of measurement 0 ⇒ m/s 3 ⇒ knot 1 ⇒ cm/s 4 ⇒ mph 2 ⇒ km/h	unsigned 16 bits	2.00	
20	Temperature unit of measurement 0 ⇒ °C 1 ⇒ °F	unsigned 16 bits	2.00	
21	Atm. pressure unit of measurement 0 ⇒ mbar (=hPa) 3 ⇒ mmH ₂ O 1 ⇒ mmHg 4 ⇒ inchH ₂ O 2 ⇒ inchHg 5 ⇒ atm	unsigned 16 bits	2.00	
22	Wind Gust intensity (x100)	unsigned 16 bits	2.20	(1),(2)
23	Wind Gust direction in degrees (x10)	unsigned 16 bits	2.20	(2)
24,25	Total amount of rainfall (x1000 if the unit of measurement is mm, x10000 if the unit of measurement is inch)	unsigned 16 bits	2.22	(3)
26,27	Partial amount of rainfall (x1000 if the unit of measurement is mm, x10000 if the unit of measurement is inch)	unsigned 16 bits	2.22	(3)
28	Rainfall rate (x10 if the unit of measurement is mm/h, x100 if the unit of measurement is inch/h)	unsigned 16 bits	2.22	
29	Rainfall unit of measurement 0 ⇒ mm 1 ⇒ inch	unsigned 16 bits	2.22	

(1) For quantities with configurable measurement unit, the measurement value is expressed in the unit set in the instrument.

(2) The Wind Gust measurement is determined by continuously calculating the wind speed averages in a time interval equal to 3 seconds, and detecting the maximum value of the calculated averages over the time elapsed between the current reading command and the previous reading command (the Wind Gust measurement is reset after each reading command).

(3) The **total** amount of rainfall is the amount measured from when the instrument is powered. The **partial** amount of rainfall is the amount measured from the last reading command. The amount of rainfall measurements are 32-bit integer values. Two consecutive 16-bit registers must be accessed to read a measurement. The register with lower address (for example the register with address 24 for the total amount of rainfall) contains the most significant bits.